

THARINDU KODIPPILI

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SUMMARY OF QUALIFICATIONS	
• 4+ years of hands-on experience building and deploying deep learning models using Python, PyTorch, and TensorFlow for time-series modelling, image/video segmentation, graph-based learning, and generative AI applications • Proven experience with distributed parallel training across multiple GPUs in PyTorch, with strong understanding of modern architectures including Transformers, graph neural networks, and diffusion-based models • Built end-to-end data pipelines in Python to automate data ingestion, processing, and dataset construction for ML training workflows • Strong software engineering foundation with production-quality code practices and experience bridging ML models with application-layer software • Domain expertise in computational modelling, simulation, and optimization – enabling the translation of complex engineering problems into effective ML solutions • Background in teaching and communicating technical concepts, with experience in undergraduate instruction covering mechanics, dynamics, and control systems •	
EDUCATION	
Doctor of Philosophy in Mechanical Engineering, University of Waterloo, Canada.	2023
Thesis: Forging Preform Design Optimization for Structural Magnesium Alloy Components Supervisors: Dr. Stephan Lambert and Dr. Arash Arami	
Masters of Applied Science in Mechanical Engineering, University of Waterloo, Canada.	2019
Thesis: Hot Forging Tool Design for Forging a Magnesium Alloy Front Lower Control Arm Supervisor: Dr. Stephan Lambert	
Courses: Mechanics of Continua, Advanced Finite Element Methods, Mechanical Metallurgy, Vehicle System Dynamics, Computational Intelligence and Neural Networks; Data and Knowledge Modelling and Analysis, Reinforcement Learning, Fundamentals of Large Language Models	

Commented [TK1]:

•Experienced with using HyperMesh, Abaqus, and nCode DesignLife; conducted strength, stiffness, and fatigue analyses on a Mg alloy suspension control arm design in accordance with MTC/Ford engineering specifications

•Industry experience in automotive R&D and manufacturing settings in both leading and supportive roles, with hands-on experience working on prototype assemblies, reviewing production and prototype drawings, participating in APQPs, and researching and designing mechanisms based on feasibility and performance

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RESEARCH EXPERIENCE

Automated Forging Preform Design Optimization Software	Present
• Ongoing development of an automated topology-based design optimization software, which relies on computational algorithms to eliminate manual CAD modelling efforts and significantly speed up design optimization • Bridges information flow between commercial CAD and FEM software • Software is written in C# using RhinoCommon and the WPF UI framework and adheres to an MVVM design pattern	
Large Deformation Simulations with MeshGraphNets	Present
• Ongoing research on applying MeshGraphNets—deep learning models for learning mesh-based simulations—to accelerate large deformation computations • Implementation of distributed parallel training in PyTorch to distribute computational load across multiple GPUs • Developed data pipelines using Python to automate simulation submission, post-processing, and mesh-to-graph conversion for building training datasets	

Development of FEM Forging Simulation Models 2023

- Developed isotropic and anisotropic material models for DEFORM-3D® to simulate the deformation of cast and extruded forms of an AZ80 Mg alloy using material data from lab-scale experiments at optimal processing conditions

Design and Analysis of Hot Forging Tool and Suspension Component Designs 2019

- Designed hot forging and pre-forming tools to forge a Magnesium alloy control arm for the 2013 Ford Fusion vehicle
- Performed thermal and mechanical analyses on tool designs to evaluate performance
- Performed strength, stiffness and fatigue life analyses of the suspension component using AZ31, ZK60, AZ80 alloy materials

INDUSTRY EXPERIENCE

Manufacturing Engineer, Bend-All Automotive, Ayr, ON 2016

- Coordinated and oversaw the flow of production to ensure timely completion of projects
- Performed inspection and verification of jig designs, machine validation testing, run-at-rates, and fault recovery testing
- Created work instructions and provided training to production members

Production Engineer (Co-op), Toyota Motor Manufacturing Canada, Cambridge, ON 2015

- Coordinated installation and mechanical design of equipment; a major project involving installation of overhead carriers for vehicle body transport was completed to the timeline by collaborating with in-house maintenance members and externals
- Designed and improved weld jigs in robot cells using Catia V5 and I-Grip software

Mechanical Engineer (Co-op), Ker-Train Research Inc., Kingston, ON 2014

- Created and updated CAD models of the GEMNI III drive train and Advanced Power Take-Off unit of the Bradley Fighting Vehicle
- Drafted machining and WEDM drawings using Inventor and AutoCAD
- Supported hardware, cooling systems, and sensor assembly of prototype designs

PUBLICATIONS

Tharindu Abesin Kodippili, Paresh Prakash, Andrew Gryguc, Behzad Behravesh, Hamid Jahed, Stephan Lambert, Bruce Williams, Fateh Fazeli. “Systematic Development of a Finite Element Model for Simulating Wrought Magnesium Alloy Forging”, International Journal of Advanced Manufacturing, 2024 (in preparation)

Tharindu Abesin Kodippili, Stephan Lambert, Arash Arami. “Multi-Objective Optimization of a Cast-Preform Shape for a Magnesium Alloy I-Beam Forging Application”, International Journal of Advance Manufacturing, 2023

Tharindu Abesin Kodippili, Stephan Lambert, Arash Arami. “Data-Driven Prediction of Forging Outcome: Effect of Preform Shape on Plastic Strain in a Magnesium Alloy Forging”, Materials Today Communications, 2022

Bruce W. Williams, **Tharindu Abesin Kodippili**, Jonathan McKinley, Stephan Lambert, Hamid Jahed. “Numerical Modeling of the Forging Response of a Magnesium Alloy Control Arm”, Mg2021 Conference Proceedings, 2021